

REZA TALANEH

Mechanical Engineer

rez.atalaneh@gmail.com • +989117440089 • Mazandaran, Iran • [Linkedin](#) • [Portfolio](#)

SUMMARY

Mechanical engineer with a MSc in Applied Mechanical Engineering and research experience in computational mechanics, finite element analysis, and smart materials. My research and industry experience have given me a strong foundation in analytical modeling, numerical simulation, and product development. I am seeking a PhD position where I can contribute to research in computational mechanics, mechanical design, advanced manufacturing, or sustainable product development while addressing real engineering challenges.

RESEARCH INTERESTS

- Applied mechanical design and structural analysis for industrial systems
- Finite element modeling of nonlinear material and structural behavior
- Smart materials and shape memory alloys; constitutive and computational modeling
- Design for manufacturability and industry-oriented product development
- Computational and applied continuum mechanics

EDUCATION

Babol Noshirvani University of Technology (THE Eng. Rank 2026: 501–600) **Babol, Mazandaran, Iran**

MSc in Applied Mechanical Engineering

Sep 2013 - Feb 2016

- Attained a GPA of 18.45/20, ranking 2nd among 24 students in the program.
- Thesis Title: "Analytical and finite element study of super elasticity behavior and shape memory effect of a Shape memory alloy thick-walled cylinder"
- Notable Courses:
Plasticity (20/20) • Theory of Plates and Shells (17.6/20) • Continuum Mechanics (18.25/20) • Finite Element Method I (19.2/20) • Advanced Numerical Computations (18.1/20)

BSc in Mechanical Engineering

Sep 2009 - Sep 2013

- Achieved a GPA of 17.33/20, securing the 2nd position out of 63 students in the program.
- Thesis Title: "Analysis of a sandwich beam with a viscoelastic core using ANSYS"
- Notable Courses:
Statics (20/20) • Mechanism Design (20/20) • Engineering Design Methods (20/20) • Mechanics of Materials • Mechanics of Composite Materials

RESEARCH EXPERIENCE

Babol Noshirvani University of Technology

Babol, Mazandaran, Iran

Graduate Researcher

Sep 2013 - Feb 2016

- Reduced the 3D Boyd–Lagoudas SMA constitutive model to plane-stress BVPs for thick-walled Ni-Ti cylinders under internal pressure and combined pressure–torsion loading, and solved the resulting coupled nonlinear equations with a custom MATLAB finite-difference solver.
- Validated results against 2 independent published solutions and 3D Abaqus FE simulations using a Fortran UMAT, characterizing martensitic phase evolution and stress redistribution under varying pressure, torsion, and temperature.
- Gained transferable skills in nonlinear constitutive modeling, numerical solution of nonlinear boundary value problems, and FEA for computational solid mechanics.

Undergraduate Researcher

May 2013 - Sep 2013

- Modeled an orthotropic composite sandwich beam with a viscoelastic core in ANSYS (SOLID186, implicit creep formulation), analyzing long-term creep strain and tip displacement under bending.

PUBLICATIONS

- Alashti, R.A., Talaneh, R., Ezami, A.H. **“Analytical and Finite Element Study of Superelasticity Behavior and Shape Memory Effect of SMA Thin-Walled Cylinders under Torsional Loading.”** *Proceedings of the 23rd International Conference on Mechanical Engineering (ISME2015), Tehran, Iran, 2015. (Oral presentation)*
- Alashti, R.A., Ezami, A.H., Talaneh, R. **“Analytical and Numerical Modeling of Pseudoelastic and Shape Memory Behavior of Shape Memory Alloys under Uniaxial Tension.”** *Proceedings of the 1st National Conference on Applied Mechanics, Azad University, Tabriz Branch, Tabriz, Iran, 2014.*

TEACHING EXPERIENCE

Babol Noshirvani University of Technology

Babol, Mazandaran, Iran

Teaching Assistant – Mechanics of Materials Laboratory

Feb 2014 - Feb 2015

- Assisted in delivering more than 20 undergraduate Mechanics of Materials laboratory sessions.

PROFESSIONAL EXPERIENCE

ABCHECK CO (Manufacturer of Agricultural & Industrial Plastic Blown Films)
Engineering Manager

Mahmudabad, Mazandaran, Iran
Oct 2017 - Present

- Leads multidisciplinary manufacturing teams, applying engineering KPIs and data-informed decision-making across product lines.

Design and Development Manager

Sep 2016 - Sep 2017

- Led engineering design, prototype development, testing, and validation.

Research and Development Supervisor

Sep 2013 - Sep 2016

- Supervised new-product development and manufacturing-process improvement.

INDEPENDENT DESIGN PROJECTS

Self-directed mechanical design projects undertaken outside of academic or employer scope:

- **Blown Film Dual-Lip Air Ring:** Reverse-engineered and modeled an industrial dual-lip air ring used in blown film extrusion, emphasizing manufacturability and mechanical functionality.
- **Two-Stage Parallel-Shaft Gearbox Design:** Complete gearbox design including transmission calculations, shaft design, bearings selection, housing development, motion verification, and manufacturing drawings.
- **Sheet Metal Enclosure:** Designed a manufacturing-ready enclosure applying sheet metal design principles, DFM, forming tools, ventilation design, and fabrication workflows.

CORE COMPETENCIES

Engineering

- Mechanical Design • CAD • DFM • FEA • Engineering KPIs • Cross-Functional Collaboration

Software

- SolidWorks • ANSYS • MATLAB • Power BI • Azure DevOps

LANGUAGES

English: **PTE Academic Overall 90/90** (CEFR C2) • German: Beginner • Persian: Native

PROFESSIONAL MEMBERSHIP

International Association of Engineers (IAENG)